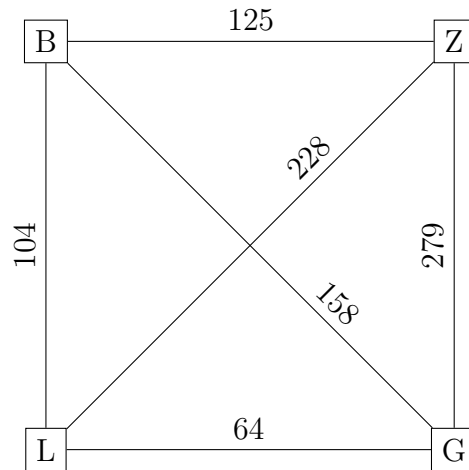


Branch and bound

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Practice quiz

Consider the traveling salesman problem with 4 cities. A salesperson living in Lausanne (L) must visit 3 customers during the day: one in Geneva (G), one in Bern (B), and one in Zurich (Z). Starting from home, she has to plan a tour, that is a sequence of customers to visit, in order to minimize the travel distance. We model the problem with a graph represented in the figure below.



Solve this problem using the branch & bound algorithm. The calculation of the lower bound can be done using the following reasoning.

Each city in the tour involves two different arcs, one to reach the city, and one to leave it. A lower bound on the length of the two arcs incident to a city is obtained by considering the two shortest arcs incident to that city. If we compute this lower bound for all nodes and add them up, we have double counted each arc (once for each of the two incident cities). Therefore, a lower bound on the total path is obtained by dividing this quantity by two.

- For Lausanne, the two shortest incident arcs have length 64 and 104.
- For Geneva, the two shortest incident arcs have length 64 and 158.
- For Bern, the two shortest incident arcs have length 104 and 125.
- For Zürich, the two shortest incident arcs have length 125 and 228.

Therefore, a lower bound on the optimal tour is

$$\frac{64 + 104 + 64 + 158 + 104 + 125 + 125 + 228}{2} = 486.$$